

HydroBoost – User Guide

1. Set the model input parameters in the Excel spreadsheets located in:
Input_Spreadsheets/Model_A (or *Model_B* / *Model_C*)
2. Chose which cases to run in the *Simulation Configuration* tab of the input Excel spreadsheet:
 - Set *Run_Flag = TRUE* for every row (case) you want to run.
 - Set the other columns (e.g., *Look_Ahead_Days*, *Interconnection Limits* *Inflow/Outflow*, *Max_Power*, *efficiencies*, etc.) per case.
 - The model reads and applies all BESS-related parameters (e.g., rated power, energy capacity, efficiencies) from this *Simulation Configuration* tab for each case.
 - During execution the model will automatically iterate only over the cases with *Run_Flag = TRUE* and will save results in separate folders named after the *Case_ID*, e.g.:
Simulation_Results/Model_C/Test 1/
Simulation_Results/Model_C/Test 3/
3. Save the updated spreadsheet in both locations:
Input_Spreadsheets/Model_A (or *Model_B* / *Model_C*)
Core_Models/HydroBoost/generated_data/Model_A (or *Model_B* / *Model_C*)
4. Refresh market data forecasts (Python). Run the Python scripts in the *Price_Forecasting* folder to refresh settings and network data. In each of *perfect_foreseight_model.py*, *mean_persistence_model.py*, and *import_data.py*, select the desired model (*Model_A*, *Model_B*, or *Model_C*) at the bottom of the script.
5. Select the model in Julia. In the Julia script *HydroBoost_Sim.jl*, select the same model (*Model_A*, *Model_B*, or *Model_C*).
6. Start a Julia REPL (terminal).
7. In the Julia REPL, run the following (example with *Model_C*):

```
import Pkg; Pkg.activate("."); Pkg.instantiate()
include("Core_Models/HydroBoost/src/HydroBoost_Sim.jl")
using .HydroBoost_Sim
result = HydroBoost_Sim.execute_HydroBoost_model("Model_C")
```

This will automatically run all cases with *Run_Flag = TRUE* and write each case's results to its own folder under *Simulation_Results/Model_C/*.
8. Review the simulation outputs in the *Simulation_Results* folder.
9. Tip: use the same model name consistently across steps 3-7.
10. For any questions, doubts, or requests, please email me at agrimaldi@anl.gov